

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 10

Level 1 — Beginner · Fractions I

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 1 — Beginner	Topic Focus	Fractions I — Parts of a Whole
Questions	20	Time Allowed	30 minutes
Total Marks	20	Worksheet No.	10 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 10 introduces fractions the friendly way — as equal parts of pizzas, ribbons, grids and groups of objects. The child reads shaded-grid fractions, names numerator and denominator, finds halves, thirds and quarters of numbers, compares unit fractions, and answers the classic 'what fraction is left?' question. Two grid diagrams make the idea visible.

Learning Objectives

- Understand a fraction as equal parts of a whole or of a group.
- Identify numerator and denominator and read fractions in words.
- Find $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ of numbers, and reverse ('half is 14, find the number').
- Compare unit fractions: same numerator, bigger denominator = smaller fraction.
- Read shaded and unshaded fractions from a picture grid.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.

SECTION A · Multiple Choice Questions · 20 Questions

Q1. A pizza is cut into 4 EQUAL pieces. Riya eats 1 piece. What fraction of the pizza did she eat?

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
 (C) $\frac{1}{4}$ (D) $\frac{4}{1}$

Q2. In the fraction $\frac{3}{8}$, which number is the **numerator**?

- (A) 3 (B) 8
 (C) 11 (D) 5

Q3. Each small square in the grid is equal. What fraction of the grid is shaded?



- (A) $\frac{3}{8}$ (B) $\frac{3}{5}$
 (C) $\frac{5}{8}$ (D) $\frac{1}{3}$

Q4. What is **half** of 18?

- (A) 6 (B) 8
 (C) 9 (D) 12

Q5. What is **one-quarter** ($\frac{1}{4}$) of 20?

- (A) 4 (B) 5
 (C) 10 (D) 15

Q6. Which fraction means '**one whole**'?

- (A) $\frac{1}{2}$ (B) $\frac{2}{4}$
 (C) $\frac{4}{4}$ (D) $\frac{1}{4}$

Q7. Which is GREATER: $\frac{1}{2}$ or $\frac{1}{4}$ of the same chocolate bar?

- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$
 (C) Both equal (D) Cannot say

Q8. A ribbon is cut into 3 EQUAL parts. What fraction is EACH part?

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
 (C) $\frac{3}{1}$ (D) $\frac{1}{4}$

Q9. What fraction of the word **SCHOOL** is made of the letter **O**?

- (A) $\frac{1}{6}$ (B) $\frac{2}{6}$
 (C) $\frac{3}{6}$ (D) $\frac{2}{5}$

Q10. Which of these shows $\frac{2}{5}$ correctly in words?

- (A) Two-fourths (B) Five-halves
 (C) Two-fifths (D) Half of five

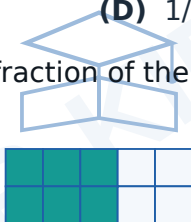
Q11. There are 12 balls in a bag: 6 are red and the rest are blue. What fraction of the balls are BLUE?

- (A) $\frac{6}{12}$ (B) $\frac{12}{6}$
 (C) $\frac{6}{6}$ (D) $\frac{1}{12}$

Q12. Which fraction is the SMALLEST?

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
 (C) $\frac{1}{5}$ (D) $\frac{1}{4}$

Q13. Each small square is equal. What fraction of the grid is NOT shaded?



- (A) $\frac{6}{10}$ (B) $\frac{4}{10}$
 (C) $\frac{3}{10}$ (D) $\frac{5}{10}$

Q14. Half of a number is 14. What is the number?

- (A) 7 (B) 21
 (C) 28 (D) 24

Q15. What is $\frac{1}{3}$ of 21?

- (A) 3 (B) 6
 (C) 7 (D) 9

Q16. Arrange from SMALLEST to GREATEST: $\frac{1}{8}$, $\frac{1}{2}$, $\frac{1}{4}$

- (A) $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$ (B) $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$
 (C) $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{2}$ (D) $\frac{1}{8}$, $\frac{1}{2}$, $\frac{1}{4}$

Q17. A glass is exactly HALF full of milk. Which fraction shows the EMPTY part of the glass?

- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$
 (C) $\frac{1}{2}$ (D) 1

Q18. Out of 10 questions, Aman answered 7 correctly. What fraction did he get WRONG?

- (A) $\frac{7}{10}$ (B) $\frac{3}{10}$
 (C) $\frac{10}{3}$ (D) $\frac{3}{7}$

Q19. Which picture fraction is the SAME as $\frac{1}{2}$: shading 2 squares out of how many?

- (A) Out of 2 (B) Out of 3
 (C) Out of 4 (D) Out of 6



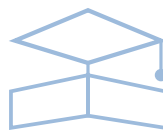
Q20. A cake is cut into 8 equal slices. Three friends eat 1 slice each. What fraction of the cake is LEFT?

(A) $\frac{3}{8}$

(B) $\frac{5}{8}$

(C) $\frac{1}{2}$

(D) $\frac{8}{5}$



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	A	A	C	B	C	A	B	B	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	A	C	B	C	C	B	C	B	C	B

Step-by-Step Solutions

Q1. Correct answer: (C)

One piece out of 4 equal pieces is $\frac{1}{4}$ (one-quarter).

Q2. Correct answer: (A)

The numerator is the TOP number, which tells how many parts are taken: **3**. The bottom number (8) is the denominator.

Q3. Correct answer: (A)

There are 8 equal squares and 3 are shaded, so the fraction is $\frac{3}{8}$.

Q4. Correct answer: (C)

Half means divide by 2: $18 \div 2 = 9$.

Q5. Correct answer: (B)

One-quarter means divide by 4: $20 \div 4 = 5$.

Q6. Correct answer: (C)

When the numerator equals the denominator, all parts are taken: $\frac{4}{4} = 1$ whole.

Q7. Correct answer: (A)

Cutting into fewer pieces makes each piece bigger. $\frac{1}{2}$ (one of 2 pieces) is greater than $\frac{1}{4}$ (one of 4 pieces).

Q8. Correct answer: (B)

One part out of 3 equal parts = $\frac{1}{3}$ (one-third).

Q9. Correct answer: (B)

SCHOOL has 6 letters, and 2 of them are O. Fraction = $\frac{2}{6}$ (which equals $\frac{1}{3}$).

Q10. Correct answer: (C)

$\frac{2}{5}$ is read as **two-fifths** — 2 parts out of 5 equal parts.

Q11. Correct answer: (A)

Blue balls = $12 - 6 = 6$. Fraction = $\frac{6}{12}$ (which is the same as $\frac{1}{2}$).

Q12. Correct answer: (C)

With the same numerator, a BIGGER denominator means smaller pieces. So $\frac{1}{5}$ is the smallest.

Q13. Correct answer: (B)

10 squares in all, 6 shaded, so $10 - 6 = 4$ are NOT shaded: $\frac{4}{10}$ (= $\frac{2}{5}$).

Q14. Correct answer: (C)

If half is 14, the whole is $14 \times 2 = 28$.

**Q15. Correct answer: (C)**

One-third means divide by 3: $21 \div 3 = 7$.

Q16. Correct answer: (B)

Same numerator $\rightarrow$ bigger denominator is smaller: $1/8 < 1/4 < 1/2$. Order: **1/8, 1/4, 1/2**.

Q17. Correct answer: (C)

If half is full, the other half is empty: **1/2**.

Q18. Correct answer: (B)

Wrong answers = $10 - 7 = 3$. Fraction wrong = **3/10**.

Q19. Correct answer: (C)

2 shaded out of 4 equal squares = $2/4 = 1/2$. (Both mean exactly half.)

Q20. Correct answer: (B)

Eaten = 3 slices out of 8. Left = $8 - 3 = 5$ slices $\rightarrow$ **5/8** of the cake.

